

18 q and $4q$ will repel each other. Thus, for the net force on q to be zero,

$$\frac{(q)(4q)}{4\pi\epsilon_0 l^2} = \frac{(q)(Q)}{4\pi\epsilon_0 (0.5l)^2}$$

$$\Rightarrow Q = q$$

Q must be of opposite charge to q so as to attract it.

$$\Rightarrow Q = -q$$

Answer: B

19 Points P and Q are at the same distance away from the positive point charge at the centre